Superconducting Connections Between NbTi Thin Films With NbN Paste

Hiroshi Takashima, Yoshiyuki Yoshida, and Mitsuho Furuse

***Abstract*—In this study, NbTi thin film-pad/NbN/NbTi thin film-pad connections are fabricated by vapor phase growth and screen-printing methods to develop applications of the NbN paste for superconducting devices, and the temperature dependence of the resistance is clarified. Consequently, two superconducting transitions are identified: a gradual decrease in resistivity in the region from 15 to 10 K, attributed to NbN, and a sharp decrease at 7.5 K, attributed to the superconducting transition of the NbTi thin film, within the accuracy of measurement by the physical properties measurement system. In the future, more accurate resistance measurements of several orders of magnitude will be required. These results are expected to be crucial for realizing three elemental technologies for superconducting thin-film device-to-device integration, consisting of a superconducting wire, superconducting crossover, and superconducting contact technology, in superconducting device circuits in a simplified manner.**

***Index Terms*—NbN, NbTi thin film-pad/NbN/NbTi thin film-pad connection, screen-printing, superconducting paste.**

I. Introduction

SILVER paste—commercially available for industrial use—is used as an electrode for screen-printed displays, flexible printed circuit boards, and piezoelectric components. Research and development regarding these applications are still underway [1], [2], [3], [4], [5]. We replaced the silver in the silver paste with NbN, which exhibits stable superconductivity, and deduced that NbN exhibits superconductivity by determining the optimal manufacturing method for the paste and the appropriate conditions for solidifying the paste into a thick film [6]. Furthermore, we confirmed that superconductivity was exhibited by creating a freestanding film and succeeded in creating a flexible film that exhibited superconductivity. Superconducting devices, such as sophisticated SQUIDs, require the development of three elemental technologies: superconducting wires, superconducting crossovers, and superconducting contact technology [7]. In this study, as an application of the developed NbN paste, we will fabricate a prototype connection structure between a known superconducting thin film and the developed NbN paste and evaluate its electrical properties to make it a candidate for connection between superconducting devices.

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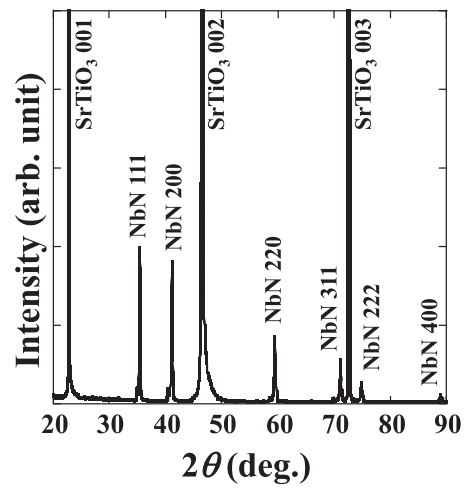

Fig. 1. XRD results of NbN paste with a thickness of 150 μm after solidification.

II. Experiments, Results, and Discussion

A. Preparation of NbN Paste and its Structural and Electrical Properties

The preparation of the NbN paste is described below. NbN superconducting metal powder (purity $\geq$ 98%, particle size: approx. 5 μm), epoxy resin (bisphenol A), solvent (diethylene glycol dibutyl ether (DGDB)), hardener (ammonium hexafluoroantimonate), and printing enhancer (silane coupling agent) were weighed. The weight ratios were as follows: NbN superconducting metal powder, 80.0%; epoxy resin, 8.0%; DGDB, 11.0%; hardener and printing enhancer, 1.0%. The weighed materials were premixed using a planetary centrifugal mixer, with the resultant mixture kneaded multiple times using a three-roll mill to prepare the final paste. Consequently, the superconducting powder, epoxy resin, curing agent, and printing enhancer are uniformly dispersed in the solvent. Next, the paste was applied to a $SrTiO_3$ (001) single-crystal substrate by screen-printing using a metal mask with a thickness of 200 μm and a 6 mm square hole. After heat curing at 180 °C for 20 min in air, the thicknesses of the pastes were 150 μm. The results indicated that the shrinkage of the NbN paste was 25%. The XRD results are shown in Fig. 1. The (001), (002), and (003) diffraction peaks of the $SrTiO_3$ (001) single-crystal substrate and those of the NbN (111), (200), (220), (311), (222), and (400) planes are confirmed, indicating that NbN is polycrystalline. No other impurity peaks were observed, thus confirming that the paste is composed of NbN alone. Solvent

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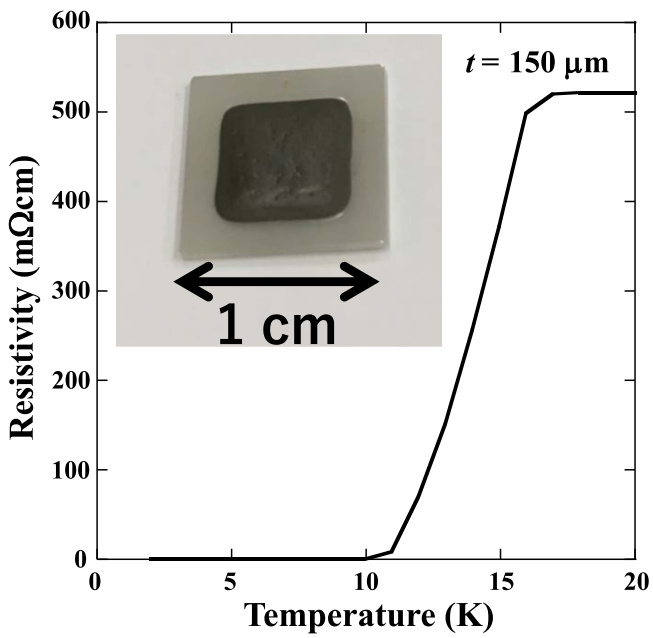

Fig. 2. Temperature dependence of resistivity of NbN paste with a thickness of 150 μm after solidification. Inset photograph shows the thick paste film prepared on $SrTiO_3$ substrate by screen-printing, exhibiting superconductivity at 10 K.

evaporated by curing, and epoxy resin, hardener, and printing enhancer remain, but are not detectable by XRD.

We have successfully obtained superconductivity in NbN films with thicknesses ranging from 80 μm to 400 μm after solidification by screen printing [6]. In this experiment, we selected an NbN paste film with a thickness of 150 μm after solidification using a metal mask with a thickness of 200 μm, which is an intermediate thickness, for the experiment.

The electrical properties of the paste with a thickness of 150 μm solidified on $SrTiO_3$(001) substrates were evaluated using a physical property measurement system (PPMS) via a four-terminal method (Fig. 2). The samples exhibited metallic electrical conductivity from 100 K to approximately 20 K, with a sharp drop observed at approximately 16 K. The resistivity becomes zero at approximately 10 K, indicating that the paste becomes a superconductor. According to previous reports [8], [9], [10], [11], the T_c of both bulk and thin-film NbN is 16 K. Our superconducting paste had a slightly lower T_c than the bulk and thin-film NbN. The detailed properties of the other film thicknesses are described elsewhere [6].

B. Preparation of NbTi-NbN Paste-NbTi Connections and Their Electrical Properties

We recently succeeded in achieving superconductivity at cryogenic temperatures by depositing NbTi-alloy thin films lattice-matched to REBCO (001) on $SrTiO_3$ (001) at room temperature [5]. Fig. 3(a) shows a schematic of the dimensions of the bridge connection structure of the NbTi thin films and NbN paste film. The NbTi film (thickness = 200 nm) was deposited on $SrTiO_3$ (001) using radio frequency (RF) magnetron sputtering with a metal mask with a gap of 1 mm between two 2 × 3.5 mm pads. Next, the NbN paste was applied to cover the gap

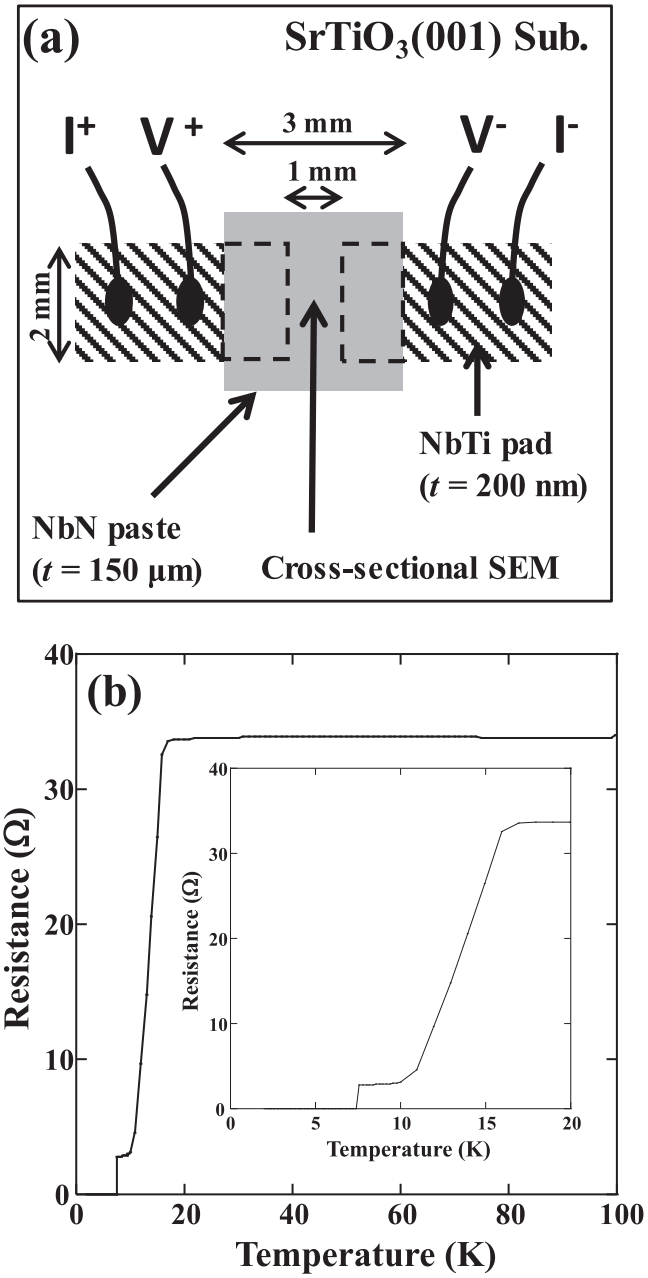

Fig. 3. (a) Schematic diagram of NbTi pad/NbN/NbTi pad connection. (b) Temperature dependence of resistivity of the NbTi pad/NbN/NbTi pad connection.

between both pads, with a part of the NbTi pads screen-printed on $SrTiO_3$ (001) using a 3 × 3 mm metal mask of 200 μm thickness. The thickness of the paste after heat curing was 150 μm. Therefore, the NbTi-pad/NbN/NbTi-pad connections were fabricated using $SrTiO_3$ (001).

The resistance was measured using a four-terminal method with a PPMS device. The lower limit of the PPMS resistance measurement is estimated to be 10 $\mu\Omega$ at a typical constant current value of 100 μA. Fig. 3(a) shows the voltage and current terminals for the four-terminal measurements. The typical results are shown in Fig. 3(b), wherein the resistance starts to decrease at approximately 15 K, with a sharp drop observed at

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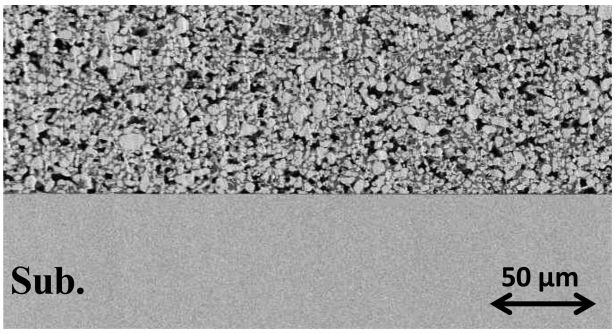

Fig. 4. Cross-sectional SEM image of the NbN paste after solidification of NbTi pad/NbN/NbTi pad connection. The NbN filler is densely packed, with many fillers in contact with adjacent fillers. (See Fig. 3(a) for measurement locations).

approximately 7.5 K. The resistance is zero below 7.5 K within the PPMS resistance measurement range. The gradual decrease in resistance from 15 K to lower temperatures is attributed to the thick NbN film because it is similar to that shown in Fig. 2. The sudden decrease in the resistance at 7.5 K is similar to the superconducting transition behavior of the NbTi thin film. The T_c of an NbTi film grown on a single-crystal substrate at room temperature was 8.5 K [12]. Here, the T_c of the NbTi thin film was reduced by approximately 1 K, possibly because of the chemical degradation of the interface containing the NbN paste during the period between screen printing and solidification, owing to the solvent in the paste. Fig. 4 shows a cross-sectional SEM image of the paste after solidification. The NbN paste is densely adhered and solidified on a $SrTiO_3$ (001) single-crystal substrate, with the cuboidal NbN fillers closely packed and many fillers in contact with adjacent fillers. The NbN particle size appears to be roughly 5 μm and is consistent with the size of the raw material. No large depletion layers are visible and they appear to be densely packed.

The NbTi-pad/NbN/NbTi-pad connection exhibited zero resistance within the temperature-dependent range of PPMS resistance measurements. The NbTi-Pad/NbN connection was expected to possess a contact resistance value below the lower limit of the PPMS resistance measurements at the contact interface. Fig. 5 shows a cross-sectional schematic of the connection structure and an equivalent circuit of the electrical resistance. The interface where the NbTi thin film is in contact with the paste may be chemically degraded, resulting in a surface-modified layer at the location indicated by the bold line in Fig. 5(a). As a result, the resistance measured by the PPMS is the resistance of the two NbTi-pads R_{NbTi}, the resistance of the NbN paste R_{NbN}, and the resistance of the two modified layers R_{SM} connected in series (Fig. 5(b)). The experimental results show that the two resistances R_{SM} of the modified layer are less than 10 $\mu\Omega$, indicating that no degradation of superconducting properties occurs on the order of this resistance, thus making it a useful candidate as a connection technique between superconducting devices.

Superconducting connections with resistance values smaller than 10^{-9} Ω are required [13], with more accurate measurements of resistance below PPMS measurement limits needed in the

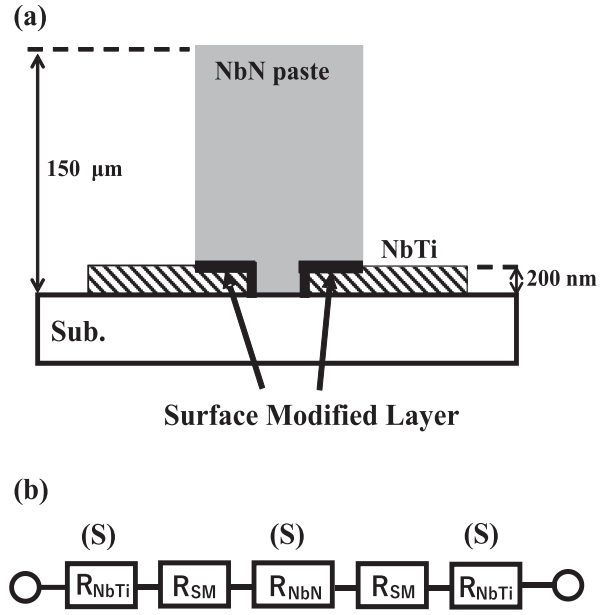

Fig. 5. (a) Cross-sectional schematic diagram of NbTi pad/NbN/NbTi pad connection. (b) NbTi pad/NbN/NbTi pad equivalent circuit of superconducting connection. R_{NbN}, R_{SM}, and R_{NbTi} are the resistance of NbN, surface-modified layer, and NbTi, respectively.

future. The proposed paste technology is easy to handle and reproducible. Furthermore, based on the results of this research, we clarified the elemental technologies consisting of superconducting wires and contacts. However, interconnect technology is necessary to wire various elements to construct electronic circuits using superconducting materials such as low T_c and high-temperature superconductors [7]. The superconducting paste technology developed in this study will be useful in addressing these issues.

III. Conclusion

NbTi thin-film-pad/NbN/NbTi thin-film pad connections were fabricated by vapor phase growth and screen-printing methods to apply the developed NbN paste to superconducting devices, with the temperature dependence of the resistivity evaluated using PPMS. Two superconducting transitions were identified: a gradual decrease in resistivity in the region from 15 to 10 K, attributed to NbN, and a sharp decrease at 7.5 K, attributed to the superconducting transition in the NbTi thin film. Below a temperature of 7.5 K, the connection resistance was found to exhibit superconductivity in the PPMS measurement range. Furthermore, although this experiment was limited to a thickness of 150 μm after solidification, it is expected to show superconductivity in the range of 80 μm to 300 μm after solidification as well, as shown in previous studies. In the future, the development of superconducting contact technology is required for the integration of multilayer superconducting devices. These results are expected to be crucial for realizing elemental technologies required for superconducting thin-film device-to-device integration and interconnect technology, consisting of superconducting wires, superconducting crossovers, and superconducting contact technology.

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